

Assignment Day 8 - Recursion

1. Factorial Using Recursion

```
#include <iostream>
using namespace std;

int factorial(int n) {
    if(n == 0 || n == 1)
        return 1;
    return n * factorial(n - 1);
}

int main() {
    int n = 5;
    cout << factorial(n);
    return 0;
}
```

Input: 5

Output: 120

2. Fibonacci Series Using Recursion

```
#include <iostream>
using namespace std;

int fibonacci(int n) {
    if(n <= 1)
        return n;
    return fibonacci(n - 1) + fibonacci(n - 2);
}

int main() {
    int n = 6;
    for(int i = 0; i < n; i++)
        cout << fibonacci(i) << " ";
    return 0;
}
```

Input: 6

Output: 0 1 1 2 3 5

3. Sum of N Numbers Using Recursion

```
#include <iostream>
using namespace std;

int sum(int n) {
    if(n == 1)
        return 1;
    return n + sum(n - 1);
}

int main() {
    int n = 5;
    cout << sum(n);
    return 0;
}
```

Input: 5

Output: 15

4. Power Function Using Recursion

```
#include <iostream>
using namespace std;

int power(int base, int exp) {
    if(exp == 0)
        return 1;
    return base * power(base, exp - 1);
}

int main() {
    cout << power(2, 5);
    return 0;
}
```

Input: 2 5

Output: 32

5. Reverse String Using Recursion

```
#include <iostream>
#include <string>
using namespace std;

void reverseString(string str) {
    if(str.length() == 0)
        return;

    cout << str[str.length()-1];
    reverseString(str.substr(0, str.length()-1));
}

int main() {
    string str = "Hello";
    reverseString(str);
    return 0;
}
```

Input: Hello

Output: olleH

6. Palindrome Check Using Recursion

```
#include <iostream>
#include <string>
using namespace std;

bool palindrome(string str, int start, int end) {
    if(start >= end)
        return true;

    if(str[start] != str[end])
        return false;

    return palindrome(str, start + 1, end - 1);
}

int main() {
    string str = "madam";

    if(palindrome(str, 0, str.length()-1))
        cout << "Palindrome";
    else
        cout << "Not Palindrome";

    return 0;
}
```

Input: madam

Output: Palindrome